

SSD_PWM usage reference

1. Parameters of PWM

1.1. Duty_cycle

- duty cycle

Echo 25 > duty_cycle means the duty cycle is 25%.

1.2. Period

- frequency

Echo 2000 > period

Represents a pwm wave with a frequency of 2kHz.

1.3. Enable/disable

- Enable
-

1.4. Polarity

- polarity.

If it is normal. Then duty_cycle=25%, which means that the proportion of high level is 25%.

If it is inverse, then vice versa.

2. Kernel configuration

2.1. Configuration of dts

```
pwm {
    compatible = "ssstar,infinity-pwm";
    reg = <0x1F003400 0x600>;
    clocks = <&CLK_xtali_12m>;
    npwm = <4>;
    pad-ctrl = <PAD_TTL0 PAD_TTL12 PAD_FUART_TX PAD_FUART_RX>;
    status = "ok";
};
```

diagram 2-1

npwm: 4 - means that pwm has 4 groups;

Pad-ctrl: Configure 4 groups of pads to be used as PWM. If you do not need to configure 4 groups of pads, the value corresponding to the pad can be set to PAD_UNKNOWN. Since gpio is shared, when gpio is used as pwm, the driver will automatically cut the pad.

Such as Display Demo Board:

infinity2m-ssc011a-s01a-display.dtsi中加入：

```
pwm {
    compatible = "ssstar,infinity-pwm";
    reg = <0x1F003400 0x600>;
```

```

clocks = <&CLK_xtali_12m>;

npwm = <2>;

pad-ctrl = <PAD_GPIO4 PAD_GPIO5>;

status = "ok"; // no available pads
};

PAD_GPIO4->PWM0 PAD_GPIO5->PWM1

nfinity2m - ssc011a - s01a - padmux.dtsi _ _

```

Revise:

```

< PAD_GPIO4 PINMUX_FOR_PWM0_MODE_3 MDRV_PUSE_PWM0 > ,

< PAD_GPIO5 PINMUX_FOR_PWM1_MODE_4 MDRV_PUSE_PWM1 > ;

```

Kernel config:

```

-[- SStar SoC platform drivers
[ ] SStar Cam drivers ----
[ ] SStar DLA driver ----
[ ] PIU Timer driver
[ ] Log Redirection
[*] Serial / UART driver
< > MS_USCLK extended sysfs interfaces for CLKS
[*] Serial Flash driver
<*> SSTAR_PWM
<*> SPI NAND
    READ (nand-qual-read) --->
    WRITE (nand-qual-write) --->
< > MS SPI driver
< > EMMC driver
<*> Mstar SD/MMC Card Interface Support
(1) Mstar SD/MMC Total Slot Number
(0) Mstar IP Select Order of SD/MMC1
(1) Mstar IP Select Order of SD/MMC2
(2) Mstar IP Select Order of SD/MMC3
↓ (+)

```

Figure 2-2

3. Hardware example

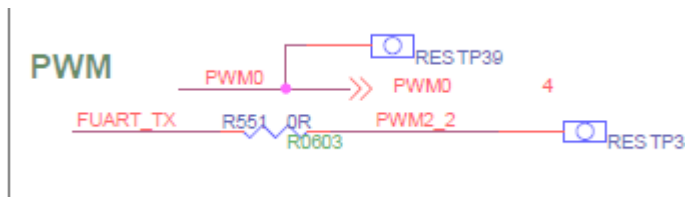


Figure 3-1

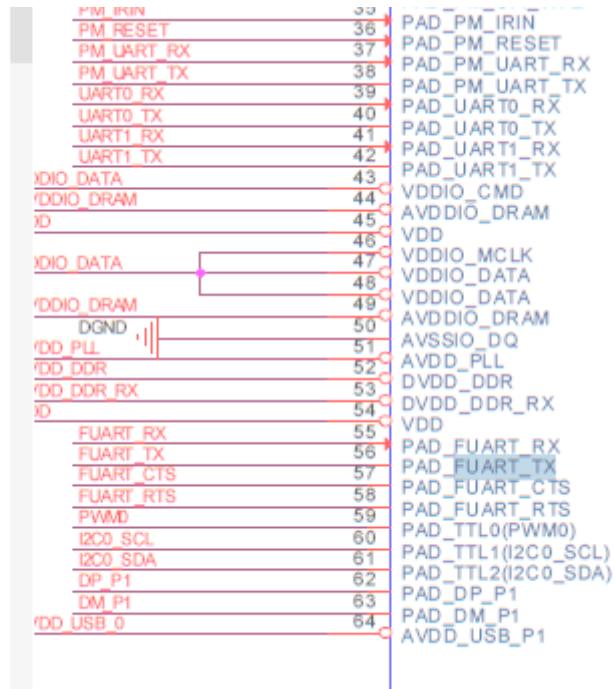
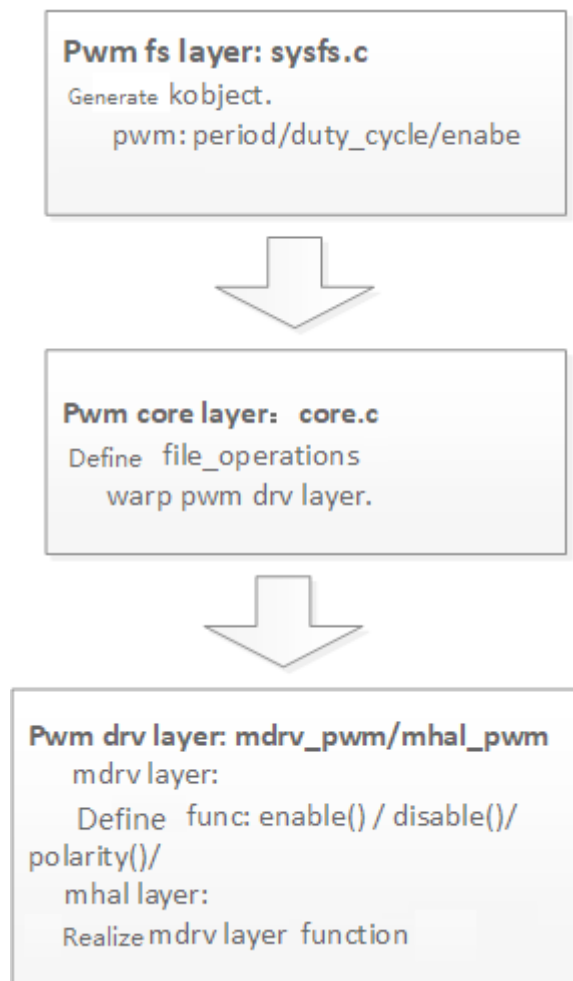


Figure 3-2

By looking at: `drivers\sstar\include\infinity2m\gpio.h`

So, just need to configure dts again.

4. PWM Architecture



Pic 4-1

5. User mode control of PWM

1. Export PWM number (例如 USB PAD_PWM0)

Command:

```
cd /sys/class/pwm/pwmchip0
```

```
echo 0 > export
```

2. Set period(frequency) / duty_cycle / polarity / enable

Command:

```
cd pwm0
```

```
echo xxxx > period
```

In our driver implementation, xxxx indicates output frequency

ex: echo 2000 > period will generate 2KHz waveform

```
echo xx > duty_cycle
```

ex: echo 25 > duty_cycle will generate 25% duty_cycle

```
echo inversed > polarity
```

Inverse output waveform, default is normal

```
echo 1 > enable
```

Enable output waveform

The corresponding user layer code only needs to operate the corresponding node, for example, the following operation duty_cycle example:

```

// set pwm duty_cycle
int SetPwmDutyCycle(int nPwm, int dutyCycle)
{
    int fd = -1;
    char pwmDev[256];
    memset(pwmDev, 0, sizeof(pwmDev));
    sprintf(pwmDev, "%s/pwm%d/duty_cycle", PWM_DEV, nPwm);

    fd = open(pwmDev, O_RDWR);

    if (fd < 0)
    {
        printf("failed to set pwm%d duty_cycle\n", nPwm);
        return -1;
    }
    else
    {
        char szDutyCycle[16] = {0};
        sprintf(szDutyCycle, "%d", dutyCycle);
        write(fd, szDutyCycle, sizeof(szDutyCycle));
        printf("set pwm%d duty_cycle: %s\n", nPwm, szDutyCycle);
        close(fd);
        return 0;
    }
} « end SetPwmDutyCycle »

```